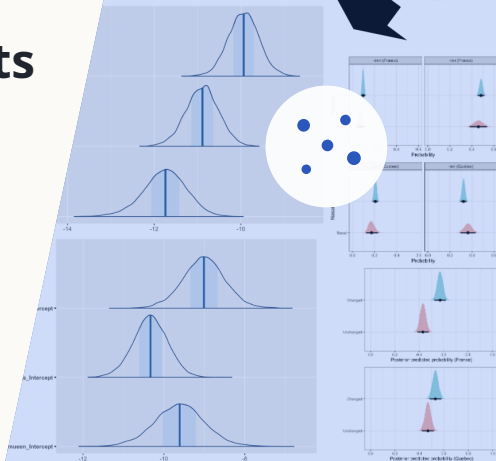


Intersecting families of phonological and extra-linguistic constraints

Evidence from French demonym formation

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ROADMAP

Agenda

- 1 Background
- 2 Method
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- 5 Conclusion
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1

Background

1 Intersecting constraint families

Intersecting constraint families (Zuraw and Hayes, 2017): Phonological outcomes are probabilistically determined by the interaction between two independent dimensions of constraints.

RATES OF VOWEL DEVOICING	constraint applying to high Vs	constraint applying to mid Vs	constraint applying to low Vs	
constraint applying to aspirated Cs	/p ^h i/ → [p ^h _i]: x%	/p ^h e/ → [p ^h _e]: y%	/p ^h a/ → [p ^h _a]: z%	
constraint applying to voiceless unaspirated Cs	/pi/ → [p _i]: w%	/pe/ → [p _e]: v%	/pa/ → [p _a]: u%	
constraint applying to voiced Cs	/bi/ → [b _i]: r%	/be/ → [b _e]: s%	/ba/ → [b _a]: r%	

Figure 1: An example of probabilistic vowel devoicing resulting from constraints on vowels (x-axis) and consonants (y-axis) (adapted from Zuraw and Hayes, 2017).

1 French demonyms

A *demonym* denotes an inhabitant of a place.

The corresponding place name is a *toponym*.

- Formed by suffixation
- Four highly frequent suffixes: *-ois*, *-ien*, *-ais*, *-éen*

(1) French demonym formation (masculine form on the left, feminine form on the right)

- | | | | |
|----|--------------------|---|--|
| a. | Paris [paʁi] | → | Parisien(ne) [paʁizjɛ̃]/ Parisienne [paʁizjɛn] |
| b. | Nice [nis] | → | Niçois [niswa]/ Niçoise [niswaz] |
| c. | Marseille [maʁsej] | → | Marseillais [maʁseje]/ Marseillaise [maʁsejez] |
| d. | Nancy [nɑ̃si] | → | Nancéen [nɑ̃seɛ̃]/ Nancéenne [nɑ̃seɛn] |

1 Factors predicting suffix choice

According to Thuilier et al. (2023), three types of factors determine suffix choice (the underlined factor is investigated in this presentation).

Morphological

- Polylexicality
- Opaque compounds
- Variation

Phonological

- Toponym length (Plénat, 2008; Roché and Plénat, 2016)
- *Final segment of toponym* (Plénat, 2008; Thuilier et al., 2023; Huygevelde et al., 2025)
- Vowel backness (Thuilier et al., 2023)

Geographical

- Types of toponym
- Local closeness (Thuilier et al., 2023)

Research question

What cues can predict suffix choice in this complex phenomenon?

Aims

- To extend Huygevelde et al. (2025)'s experiment using a broader range of stimuli
- To incorporate a geographical factor into the experiment
 - Tested by comparing two geographically distant participant groups
- To model phonological and extralinguistic variables
- To provide a theoretical account

2

Method



2 Current experiment: overview

- **Type** Four-alternative forced-choice (4AFC) experiment
- **Set-up** jsPsych (De Leeuw, 2015) / GitHub Pages / DataPipe(De Leeuw, 2023)
- **Data collection** Data automatically stored in an OSF repository via DataPipe (De Leeuw, 2023)

Two participant groups

Metropolitan French (MF)

- $N = 27$, gender: $F = 22, M = 5$; age: $M_{age} = 40.33(sd = 16.48)$
- **Recruitment**: RISC (UAR 3332 CNRS; <https://www.risc.cnrs.fr/expescience>)

Quebec French (QF)

- $N = 26$, gender: $F = 8, M = 18$; age: $M_{age} = 36.88(sd = 10.24)$
- **Recruitment**: Prolific (<https://www.prolific.com/>; Palan and Schitter, 2018)

2 Stimuli and carrier sentence I

Stimuli

Jumeu_e [ʒymœ _____]

{b, d, f, g, j, k, l, m, n, p, ʁ, s, ʃ, t, v, z, ʒ}

- ***JumeuXe*** was assumed to be phonologically unmarked in French.
- Of the 17 stimuli, 9 were *masculine* and 8 were *feminine*.

Carrier

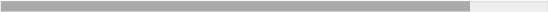
JumeuXe est une ville (dans la région [REGION NAME]). Paul vient de JumeuXe. Il est __.

- Participants were asked to fill in the blank.
- **Presentation:** Stimuli were presented both orthographically and auditorily; the audio was generated using the Python library `pyttsx3`.

1. Informed consent
2. Instruction
3. Comprehension test
 - *IF ALL ANSWERS WERE CORRECT:* Proceed to the next task
 - *OTHERWISE:* Return to the instructions
4. Practice task with existing toponyms (*Paris, Europe*)
5. **Task 1: 4AFC *without any geographical information***
6. **Task 2: 4AFC *with information about the region to which the place belongs***



Figure 2: The four regions labelled in Task 2, each associated with a different linguistic group.

Etat d'avancement 

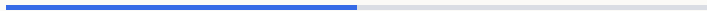
Écoutez l'enregistrement et choisissez parmi les propositions celle qui convient pour remplir l'espace vide.
Veuillez choisir l'option la plus naturelle selon vous parmi les choix.

Jumeugue est une ville dans la région Alsace. Julie vient de Jumeugue. Elle est ____.

Figure 3: An example from Task 2. The geographical information ("*dans la région Alsace*") was not presented in Task 1.

3

Results



Bayesian mixed-effects multinomial logistic regression

- Implemented in R (R Core Team, 2025) using `brms` (Bürkner, 2017)
- Predicts the probability of *selecting each suffix* from several predictors
 - **Reference category:** *-ois* for MF and *-ien* for QF, because each was the most frequent suffix in its respective group

Model: $RESPONSE \sim \text{segment} + \text{geo_info} + \text{genre} + \text{offset}(\log(\text{rt})) + (1 \mid \text{participant})$

- **Fixed predictors:** **Final segment** (17 levels; reference: *global mean*), presence of a **geographical label**, and **grammatical gender** (masculine or feminine)
- **Offset:** Log-transformed reaction time
- **Random intercept:** Participant

Prior distributions and MCMC settings

- Weakly informative priors (Lemoine, 2019): $b \sim \mathcal{N}(0, 1)$, Intercept $\sim t_3(0, 2.5)$, $\text{sd} \sim t_3(0, 2.5)$
- Valid MCMC chains: (4,000 iterations – 2,000 warm-up iterations) $\times$ 4 chains = 8,000 post-warm-up samples

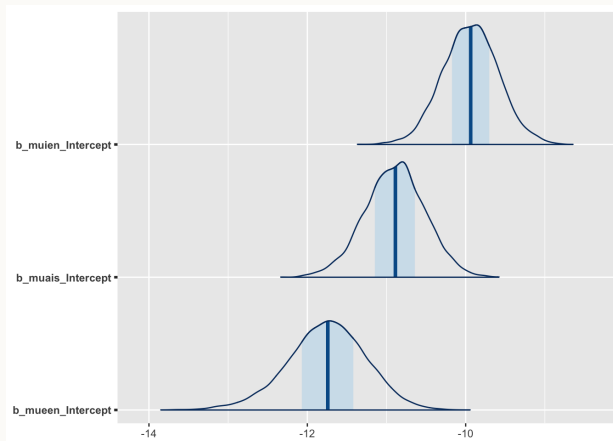


Figure 4: Intervals and densities for the estimated mean selection probability of each suffix relative to *-ois*.

Table 1: Suffix-selection estimates relative to *-ois*^a

Predictor	β	Error	95% CI	pd
<i>-ien</i> _intercept	-9.89	0.36	[-10.59 – -9.20]	≈100%
<i>-ais</i> _intercept	-10.90	0.38	[-11.66 – -10.16]	≈100%
<i>-éen</i> _intercept	-11.77	0.50	[-12.80 – -10.84]	≈100%

- Suffix selection tendency (*lexical propensity*): ***-ois* > *-ien* > *-ais* > *-éen***
- Consistent with patterns in attested demonyms reported by Thuilier et al. (2023)

^a **Estimate** (β): estimated effect size. **95% credible interval** (95% CI): range of plausible parameter values. **Probability of direction** (pd): probability that the effect has the reported direction.

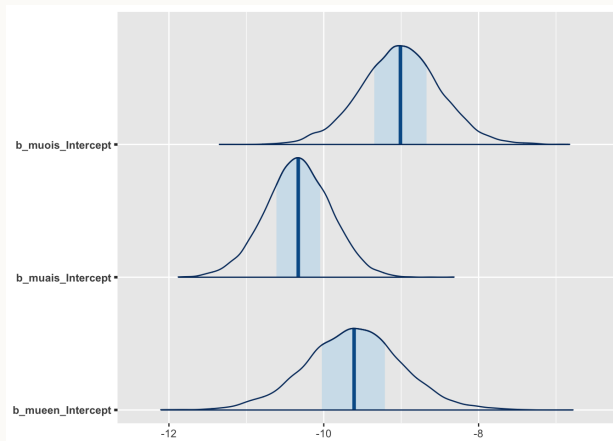


Figure 5: Intervals and densities for the estimated mean selection probability of each suffix relative to *-ien*.

Table 2: Suffix-selection estimates relative to *-ien*

Predictor	β	Error	95% CI	pd
<i>-ois_intercept</i>	-8.99	0.53	[-10.00 – -7.93]	≈100%
<i>-éen_intercept</i>	-9.66	0.61	[-10.89 – -8.47]	≈100%
<i>-ais_intercept</i>	-10.34	0.43	[-11.16 – -9.49]	≈100%

- Lexical propensity: *-ien* > *-ois* > *-éen* > *-ais*
- Different from the MF result and from attested demonym data
- Before estimating the other predictors, **grammatical gender did not affect suffix choice in either variety**. (MF: $\beta = 0.86$, QF: $\beta = -0.32$)

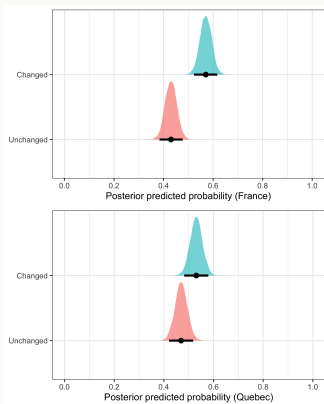


Figure 6: Predicted probability of changing the response between tasks for the same final segment.

- The presence of geographical information did not affect suffix choice.
 - MF** *geo_infoY* vs. *N*: $\beta = -0.67$, 95% CI = $[-1.73 - 0.39]$
 - QF** *geo_infoY* vs. *N*: $\beta = 0.86$, 95% CI = $[-0.68 - 2.41]$

Table 3: Both pools showed little evidence of response change.

Condition	95% CI	Post.Prob
MF: Changed	[0.49 – 0.59]	54.3%
MF: Unchanged	[0.41 – 0.51]	45.7%
QF: Changed	[0.49 – 0.58]	53.4%
QF: Unchanged	[0.42 – 0.51]	46.6%

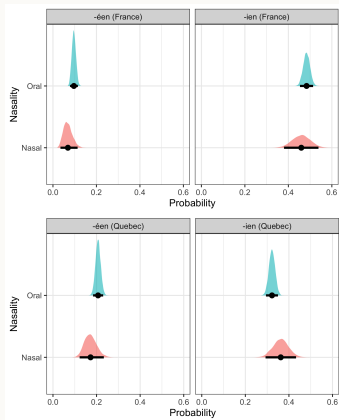


Figure 7: Probability of choosing nasal suffixes in the two varieties.

Table 4: Conditional probability of the OCP effect.

Condition	Odds Ratio	95% CI	Post.Prob
MF			
$\frac{\Pr(ien +nasal)}{\Pr(ien -nasal)}$	0.77	[0.56 – 1.01]	97.1%
$\frac{\Pr(een +nasal)}{\Pr(een -nasal)}$	0.73	[0.37 – 1.26]	87.0%
QF			
$\frac{\Pr(ien +nasal)}{\Pr(ien -nasal)}$	1.14	[0.90 – 1.41]	13.1%
$\frac{\Pr(een +nasal)}{\Pr(een -nasal)}$	0.80	[0.56 – 1.11]	90.2%

- A **modest inhibitory effect** of nasal clash was observed, except for QF -ien.
 - Except for QF -ien, nasal-final bases reduced the odds of nasal suffixes by factors of 0.7–0.8.

- **Strong effect** of *lexical propensity*
 - MF *-ois* > *-ien* > *-ais* > *-éen*
 - QF *-ien* > *-ois* > *-éen* > *-ais*
- **Negligible effect** of *geographical knowledge*
 - The two participant groups performed similarly despite their different geographical knowledge.
- **Modest inhibitory effect** of *nasal clash*.
 - Nasal suffixes were disfavoured after nasal-final bases, with odds ratios of 0.7–0.8.
 - **Exception: QF *-ien***
Possibly due to the high lexical propensity of *-ien* in QF.

4

MaxEnt analysis

Maximum Entropy Harmonic Grammar (MaxEnt: Goldwater and Johnson, 2003; Hayes and Wilson, 2008): A probabilistic approach to constraint-based phonology.

- Constraints are weighted, not ranked

$$\arg \max(\vec{w}) \sum_{i=1}^K \ln \Pr(y_i | x_i, \vec{w}) \quad (1)$$

$$H = \sum_{i=1}^K w_i C_i \quad (2)$$

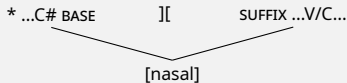
$$\Pr = \frac{e^{-H}}{\sum_{i=1}^N e^{-H}} \quad (3)$$

Symbols

- w : Constraint weight
- x : Candidate
- y : Input
- H : Harmony
- C : Number of violations

(2) **FAMILY 1: Phonological**

- a. *...[nasal]][...[nasal]...: Assign one violation to a candidate in which the base-final and suffix-initial segments are both associated with [nasal].

(3) **FAMILY 2: Gradually learned lexical propensity** (cf. USE-X: Zuraw and Hayes, 2017)

- a. USE-*ois*: Assign one violation to a candidate which does not contain *-ois*
- b. USE-*ien*: Assign one violation to a candidate which does not contain *-ien*
- ...

Table 5: Simulation of toponyms ending in a nasal: $\loglik = -131.26$

w : weight; violations are indicated numerically; $H = \sum_{i=1}^{K=5} n(\text{"1''}) \times n(\text{const})$; Pr(obs): observed probability; Pr(exp): predicted probability.

Inp: ...{m, n}e + suf	USE-ois $w = 0.87$	*nasal][nasal $w = 1.09$	USE-ien $w = 1.27$	USE-ais $w = 0.22$	USE-éen $w = 0.00$	H	e^{-H}	Pr(obs)	Pr(exp)
...{m, n}ois			1	1	1	0.13	0.87	49.5%	49.5%
...{m, n}ais	1		1		1	0.78	0.45	25.7%	25.7%
...{m, n}ien	1	1		1	1	0.82	0.44	24.8%	24.8%
...{m, n}éen	1	1	1		1	3.46	0.03	6.5%	6.5%

- Weight optimization and probability calculations were conducted using `maxent.ot` (Mayer et al., 2024)
- The close fit between observed and predicted probabilities indicates good model convergence
- Two families intersect
- Despite lexical propensity, *-ais* is favoured over *-ien* because of *nasal][nasal

Table 6: Simulation of toponyms ending in a nasal: $\loglik = -135.96$

w : weight; violations are indicated numerically; $H = \sum_{i=1}^{K=5} n("1") \times n(const)$; Pr(obs): observed probability; Pr(exp): predicted probability.

Inp: ...{m, n}e + suf	USE-ois $w = 0.96$	*nasal][nasal $w < 0.01$	USE-ien $w = 0.79$	USE-ais $w = 0.00$	USE-éen $w = 0.13$	H	e^{-H}	Pr(obs)	Pr(exp)
...{m, n}ois			1	1	1	0.13	0.87	37.5%	37.5%
...{m, n}ais	1		1		1	0.78	0.45	14.4%	14.4%
...{m, n}ien	1	1		1	1	0.82	0.44	31.7%	31.7%
...{m, n}éen	1	1	1	1		3.46	0.03	16.3%	16.3%

- **The weight of *nasal][nasal is near zero**
 - Consequently, *-ien* and *-éen* are favoured.

Table 7: Differences in constraint weights between MF and QF

Variant	USE-ois	USE-ien	USE-ais	USE-éen	*nasal][nasal
MF	0.87	1.27	0.22	0.00	1.09
QF	0.96	0.79	0.00	0.13	<0.01

- Weights are not directly comparable across fitted models (these models are not intended for typological comparison)
- The largest cross-variety difference is the weight of *nasal][nasal
- Nasal clash creates an intersection that yields different outcome probabilities
 - A similar nasal-clash effect: **Japanese “paired names”** (Kumagai and Kawahara, 2018)
 - Nasal clash also operates weakly across word boundaries, disfavours adjacent nasals: e.g., *mana#kana*, **kana#mana*, *miu#mima*, **mima#miu*, *mima#hina*, **hina#mima*

5

Conclusion



- Lexical propensity plays a strong role in predicting suffix choice.
 - Native French speakers may use rules of thumb when forming demonyms
- The interaction between lexical propensity and probabilistic phonology was confirmed
- Constraint weights:
 MF: $w(\text{USE-ois}) \gg w(*\dots[\text{nasal}])[\dots[\text{nasal}]\dots] \gg w(\text{USE-ien}) \gg w(\text{USE-ais}) \gg w(\text{USE-éen})$
 QF: $w(\text{USE-ois}) \gg w(\text{USE-ien}) \gg w(\text{USE-éen}) \gg w(*\dots[\text{nasal}])[\dots[\text{nasal}]\dots] \gg w(\text{USE-ais})$
- Differences in this interaction yield different probabilities of phonological outcomes

5 Future directions: demonym frequency and variation

- Highly frequent demonyms such as *Parisien* and *Européen* are nearly categorical
- Lexical frequency may affect language use (cf. usage-based models: Pierrehumbert, 2001; Todd et al., 2019).
- Although demonym formation is probabilistic, one candidate may dominate when predictability is high

Table 8: Practice-task data for *Paris-SUF*: $\loglik = -135.96$

Inp: Paris + suf	USE-ois $w = 0.00$	*nasal][nasal $w = 0.00$	USE-ien $w = 20.25$	USE-ais $w = 0.00$	USE-éen $w = 0.00$	H	e^{-H}	Pr(obs)	Pr(exp)
Parisien	1			1	1	0	1	100%	>99.99%
Parisois			1	1	1	20.25	<0.01	0%	<0.01%
Parisais	1		1		1	20.25	<0.01	0%	<0.01%
Pariséen	1		1	1		20.25	<0.01	0%	<0.01%

6

References

References

- Bürkner, Paul-Christian. 2017. "brms: An R package for Bayesian multilevel models using Stan". *Journal of statistical software*, 80, pp. 1–28.
- De Leeuw, Joshua R. 2015. "jsPsych: A JavaScript library for creating behavioral experiments in a Web browser". *Behavior research methods*, 47(1), pp. 1–12.
- De Leeuw, Joshua R. 2023. "DataPipe: Born-open data collection for online experiments". *Behavior Research Methods*, 56(3), pp. 2499–2506. doi: 10.3758/s13428-023-02161-x.
- Goldwater, Sharon and Mark Johnson. 2003. "Learning OT constraint rankings using a maximum entropy model", pp. 111–120.
- Hayes, Bruce and Colin Wilson. 2008. "A maximum entropy model of phonotactics and phonotactic learning". *Linguistic inquiry*, 39(3), pp. 379–440.
- Huygevelde, Marie, Ridvan Kayirici, Olivier Bonami, and Barbara Hemforth. 2025. "Affix rivalry in French demonyms: an experimental approach". *Lexique*, Numéro special, pp. 31–51.
- Kumagai, Gakuji and Shigeto Kawahara. 2018. "Stochastic phonological knowledge and word formation in Japanese". *GENGO KENKYU (Journal of the Linguistic Society of Japan)*, 153, pp. 57–83.
- Lemoine, Nathan P. 2019. "Moving beyond noninformative priors: why and how to choose weakly informative priors in Bayesian analyses". *Oikos*, 128(7), pp. 912–928. doi: 10.1111/oik.05985.
- Mayer, Connor, Adeline Tan, and Kie Ross Zuraw. 2024. "Introducing maxent. ot: an R package for Maximum Entropy constraint grammars". *Phonological Data and Analysis*, 6(4), pp. 1–44.
- Palan, Stefan and Christian Schitter. 2018. "Prolific.ac—A subject pool for online experiments". *Journal of Behavioral and Experimental Finance*, 17, pp. 22–27. doi: <https://doi.org/10.1016/j.jbef.2017.12.004>.
- Pierrehumbert, Janet B. 2001. "Exemplar dynamics: Word frequency, lenition and contrast", vol. 45, pp. 137–158. doi: 10.1075/tsl.45.08pie.
- Plénat, Marc. 2008. "Quelques considérations sur la formation des gentilés". *La raison morphologique: Hommage à la mémoire de Danielle Corbin*, pp. 155–174.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*.
- Roché, Michel and Marc Plénat. 2016. "De l'harmonie dans la construction des mots du français". *Actes du cinquième Congrès mondiale de linguistique française*.
- Thuilier, Juliette, Delphine Tribout, and Marine Wauquier. 2023. "Affix rivalry in French demonym formation: The role of linguistic and non-linguistic parameters". *Word Structure*, 16(1), pp. 115–146. doi: 10.3366/word.2023.0223.
- Todd, Simon, Janet B. Pierrehumbert, and Jennifer Hay. 2019. "Word frequency effects in sound change as a consequence of perceptual asymmetries: An exemplar-based model". *Cognition*, 185, pp. 1–20. doi: 10.1016/j.cognition.2019.01.004.
- Zuraw, Kie and Bruce Hayes. 2017. "Intersecting constraint families: an argument for Harmonic Grammar". *Language*, 93(3), pp. 497–548.

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Questions and comments are welcome in both
Japanese and English

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My website

